

NTUNG NGELA LANDON

Cybersecurity Researcher — AI Security — Biometric Security

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RESEARCH PROFILE

Cybersecurity researcher focused on the security, fairness, robustness, and generalization of AI-enabled biometric systems. My research centers on face Presentation Attack Detection (PAD), with particular emphasis on demographic disparities affecting African populations and the reliability of fairness claims across architectures, datasets, and experimental conditions. My research trajectory progresses from developing fairness-aware PAD interventions, to investigating architectural sources of demographic disparity, to evaluating whether observed fairness properties remain stable under distribution shift and random-seed variation. I have contributed as a lead researcher to two published papers and two additional manuscripts currently under review.

RESEARCH CONTRIBUTIONS

- **Fairness-aware biometric security:** Developed and evaluated a lightweight PAD methodology combining Local Binary Patterns, ethnicity-aware preprocessing, group-specific threshold optimization, and statistical fairness analysis to reduce demographic disparities in face PAD.
- **Architectural bias in biometric AI:** Led a comparative investigation of convolutional and Vision Transformer architectures to examine whether model architecture contributes to demographic disparities and cross-demographic generalization in face PAD.
- **Fairness under distribution shift:** Extended the research program to evaluate whether fairness measurements obtained from a single dataset and training run remain reliable under cross-dataset distribution shift, controlled ablations, and random-seed replication.
- **Experimental rigor and reproducibility:** Applied cross-dataset evaluation, Single-Source Generalization, Leave-One-Dataset-Out evaluation, five-seed replication, controlled ablation studies, and statistical analysis to distinguish genuine fairness effects from experimental variability.

RESEARCH INTERESTS

Biometric Security; Face Presentation Attack Detection; AI Security; Fairness and Responsible AI; Trustworthy AI; Machine Learning Security; Computer Vision; Robustness and Generalization of AI Systems; Demographic Bias in AI; Security of AI-Enabled Authentication Systems.

EDUCATION

Carnegie Mellon University

Master of Science in Information Technology – Cybersecurity

GPA: 3.64/4.00

Pittsburgh, USA

August 2024 – December 2025

- Final Practicum: *Cross-Platform Automated Malware Analysis Pipeline.*
- Graduate research in biometric security, AI security, machine learning, and fairness-aware computer vision.

University of Buea

Bachelor of Engineering in Computer Engineering

GPA: 3.51/4.00

Buea, Cameroon

October 2019 – June 2023

- Final-Year Project: *Design and Implementation of a Cognitive-Based IoT Fetal Movement Monitoring System.*

RESEARCH EXPERIENCE

Carnegie Mellon University

Research Associate

March 2026 – March 2027

- Lead research investigating the robustness and reproducibility of demographic fairness in face Presentation Attack Detection (PAD) under distribution shift.
- Extend prior fairness-aware and architecture-focused PAD research to determine whether reported fairness outcomes remain stable across datasets, evaluation protocols, and random seeds.
- Evaluate a two-branch PAD framework across the CASIA-SURF CeFA, OULU-NPU, and SiW-M benchmarks using Single-Source Generalization (SSG) and Leave-One-Dataset-Out (LODO) protocols.
- Design and conduct five-seed replication experiments and controlled ablation studies to distinguish genuine fairness effects from random-seed and evaluation variability.
- Observe stable aggregate discrimination under cross-dataset shift, with mean AUCs of 76.8% for SSG and 76.7% for LODO, while demographic fairness gaps remain substantially larger and less stable.
- Measure a mean best-to-worst treated-group fairness gap of 30.6 percentage points on the ethnicity-labelled CeFA test set across five seeds, with observed gaps ranging from 23 to 39 percentage points.
- Demonstrate that no individual design choice, including the second model branch, multi-source training, or ethnicity-aware preprocessing, produced a statistically significant change in the fairness gap after accounting for seed variance.
- Manuscript: *Fairness Under Distribution Shift in Face Presentation Attack Detection: A Cross-Dataset and Ablation-Based Analysis* (under review).
- Contribute to a separate Microsoft-funded cybersecurity research project focused on malware analysis and security evaluation.

Carnegie Mellon University

Research Assistant

August 2025 – December 2025

- Led an empirical investigation into whether model architecture contributes to demographic disparities in face Presentation Attack Detection.
- Designed and implemented comparative experiments evaluating Multimodal ViT-Tiny, ResNet18, and pretrained DeiT-S architectures on the CASIA-SURF CeFA dataset.
- Evaluated models across African, East Asian, and zero-shot Central Asian demographic groups to investigate demographic fairness and cross-demographic generalization.
- Conducted PAD evaluation using accuracy, APCER, BPCER, ACER, and Equal Error Rate (EER).
- Demonstrated that pretrained DeiT-S achieved 97.27% overall accuracy and 0.86% EER, reducing the African–East Asian ACER gap to 0.13%.
- Demonstrated improved zero-shot generalization to Central Asian subjects, with DeiT-S achieving 2.89% BPCER compared with 10.44% for ResNet18.
- Led manuscript development and research interpretation resulting in the paper: *Architectural Bias in Face Presentation Attack Detection: A Comparative Study of Vision Transformers and Convolutional Neural Networks*.

Carnegie Mellon University

Research Intern

June 2025 – August 2025

- Initiated research investigating demographic fairness in face Presentation Attack Detection for underrepresented African populations.
- Developed a lightweight PAD pipeline using Local Binary Patterns (LBP), ethnicity-aware preprocessing, and group-specific decision threshold optimization.
- Designed and implemented adaptive preprocessing using CLAHE and gamma correction to address illumination and image-quality disparities across skin tones.

- Implemented group-specific decision thresholds using Equal Error Rate (EER) minimization.
- Designed the fairness evaluation methodology using coefficient of variation, McNemar’s statistical testing, and bootstrap confidence intervals.
- Conducted dataset preprocessing, feature extraction, experimentation, validation, visualization, and analysis using the CASIA-SURF CeFA dataset.
- Achieved 95.12% accuracy and 98.55% AUC, reducing the African–East Asian accuracy gap from 3.07% to 0.75%.
- Contributed to manuscript development and publication of: *Fairness-Aware Face Presentation Attack Detection Using Local Binary Patterns: Bridging Skin Tone Bias in Biometric Systems*.

PUBLICATIONS

Peer-Reviewed Journal Publication

1. Jema David Ndibwile, **Ntung Ngela Landon**, and Floride Tuyisenge. “Fairness-Aware Face Presentation Attack Detection Using Local Binary Patterns: Bridging Skin Tone Bias in Biometric Systems.” *Journal of Cybersecurity and Privacy*, 6(1), 2026. doi.org/10.3390/jcp6010012.

Conference Paper / Preprint

2. **Ntung Ngela Landon**, Floride Tuyisenge, and Jema David Ndibwile. “Architectural Bias in Face Presentation Attack Detection: A Comparative Study of Vision Transformers and Convolutional Neural Networks.” *arXiv:2606.18510*, 2026. doi.org/10.48550/arXiv.2606.18510.

Manuscripts Under Review

3. **Ntung Ngela Landon**, Floride Tuyisenge, Remy Dukundane, Emmanuel Iduh, and Jema David Ndibwile. “Fairness Under Distribution Shift in Face Presentation Attack Detection: A Cross-Dataset and Ablation-Based Analysis.” Manuscript under review.
4. Jema David Ndibwile, **Ntung Ngela Landon**, and Lunodzo Mwinuka. “Adversarial Artificial Intelligence Threats in Smart Energy Grids: When Learning Systems Learn from the Attacker.” Manuscript under review.

RESEARCH PRESENTATIONS

- “**Architectural Bias in Face Presentation Attack Detection: A Comparative Study of Vision Transformers and Convolutional Neural Networks.**” Presented at the *10th International Conference on Cryptography, Security and Privacy (CSP 2026)*, Sapporo, Japan, April 2026.

TEACHING EXPERIENCE

Carnegie Mellon University

Teaching Assistant – CMU Africa Bridge Program

Kigali, Rwanda

January 2025 – April 2025

- Supported undergraduate students preparing for graduate-level study in computing and technology through the CMU Africa Bridge Program.
- Assisted students with technical learning activities, problem solving, and foundational concepts required for successful transition into graduate education.
- Supported instructional delivery, student engagement, and academic learning activities.

PROFESSIONAL EXPERIENCE

International Telecommunication Union (ITU), Switzerland

Remote: AI, Data Management, and Cybersecurity Intern

July 2025 – November 2025

- Conducted hands-on security assessment of an organizational platform using an existing penetration-testing report as a baseline for further vulnerability investigation.

- Independently explored the platform to identify additional potential vulnerabilities and security weaknesses.
- Investigated the Digital Object Architecture (DOA) and Advanced Information Management System (AIMS).
- Examined the interconnectedness of artificial intelligence, data management, and data security in AI-enabled information systems.

Reseaux et Telecommunications (Resotel), Cameroon

Networks and Telecommunications Intern

September 2022 – February 2023

- Supported installation and initial configuration of surveillance camera systems to strengthen organizational security infrastructure.
- Maintained optical fiber connections to ensure reliable and secure data transmission.
- Deployed a Synology Network Attached Storage (NAS) server to improve secure data storage and accessibility.
- Implemented the 3CX communication platform to improve internal organizational communication.

SELECTED RESEARCH AND TECHNICAL PROJECTS

Cross-Platform Automated Malware Analysis Pipeline

MSc Final Practicum – Carnegie Mellon University

- Designed and implemented an automated malware-analysis pipeline capable of handling both Windows and Linux malware.
- Integrated a CAPE Sandbox environment with dedicated Windows and Ubuntu virtual machines to execute malware samples in controlled environments.
- Designed an automated classification workflow that identifies whether a malware sample targets Windows or Linux and routes the sample to the appropriate analysis environment.
- Developed an ingestion architecture connecting a honeypot network and malware sample collection to event-hash processing and ingestion scripts that channel samples to the CAPE analysis host.
- Automated behavioral analysis and generated structured database reports containing analysis results.
- Integrated the resulting analysis pipeline with the SOC SIEM of the CMU-Upanzi network, enabling security monitoring and downstream analysis of malware activity.

IoT-Based Fetal Health Monitoring System

BEng Final-Year Project – University of Buea

- Designed and implemented an IoT-based system for monitoring fetal movement and health indicators.
- Developed signal-processing algorithms for filtering and analyzing fetal health signals.
- Integrated wireless communication modules for transmitting health data to remote monitoring stations.
- Developed interfaces for visualizing real-time fetal heart rate and temperature measurements.

RESEARCH METHODS

Experimental Design: Controlled Experiments; Ablation Studies; Cross-Dataset Evaluation; Single-Source Generalization; Leave-One-Dataset-Out Evaluation; Multi-Seed Replication; Reproducibility Analysis.

Biometric Evaluation: APCER; BPCER; ACER; EER; ROC/AUC Analysis; Demographic Performance Gap Analysis.

Statistical Analysis: Bootstrap Confidence Intervals; McNemar’s Test; Coefficient of Variation; Statistical Significance Testing.

Machine Learning: CNNs; Vision Transformers; Transfer Learning; Feature Extraction; Classification; Model Evaluation; Fairness-Aware Machine Learning.

TECHNICAL SKILLS

Cybersecurity: Penetration Testing; Ethical Hacking; Vulnerability Assessment; Malware Analysis; Network Security; Wireless Security; Biometric Security; Presentation Attack Detection.

AI and Machine Learning: Python; TensorFlow; OpenCV; scikit-learn; NumPy; Pandas; Matplotlib; Computer Vision; Deep Learning.

Security and Networking Tools: Kali Linux; Wireshark; Burp Suite; Metasploit; Nmap; Nikto; Hydra; CAPE Sandbox.

Systems and Infrastructure: Linux; Windows; Ubuntu; Virtualization; Optical Fiber Networks; NAS; 3CX; SIEM.

Development: Git/GitHub; C/C++; JavaScript; Arduino IDE.

AWARDS AND HONORS

- **Future Africa Leaders Award**, 2024.
- **Leadership Initiative Award**, 2020.
- **Afretec Inclusion Award**, Women in Tech Club, CMU.

LEADERSHIP AND SERVICE

Events Manager, Women in Tech Club, Carnegie Mellon University.

Partnership and Sponsorship Lead, Africa Forge Challenge 2.0, Future Africa Leaders Foundation.

Future Africa Leaders Foundation Ambassador.

LANGUAGES

English: Fluent / Native

French: Intermediate